
Ground water level prediction using artificial neural network

Noor-E-Ashmaul Husna*

Institute of Water and Flood Management,
Bangladesh University of Engineering and Technology, Dhaka
Email: husna_cee@yahoo.com

*Corresponding author

Sheikh Hefzul Bari

Dept. of Civil Engineering,
Leading University,
Sylhet, Bangladesh
Email: bari_cee1989@yahoo.com

Md. Manjurul Hussain

Dept. of Civil and Environmental Engineering,
Shahjalal University of Science and Technology,
Sylhet, Bangladesh
Email: mmhs013@gmail.com

Md. Tauhid Ur-Rahman

Department of Civil Engineering,
Military Institute of Science and Technology,
Dhaka, Bangladesh
Email: tauhid_cee@yahoo.com

Mashrekur Rahman

Institute of Water and Flood Management,
Bangladesh University of Engineering and Technology, Dhaka
Email: viki.mashrekur2@gmail.com

Abstract: In this paper, the feedforward neural network was used to predict the groundwater level at Chandpur District of Bangladesh. Levenberg-Marquardt (LM) algorithm was used as network training algorithm and sigmoid function as the transfer function. Weekly groundwater level data of six measuring wells from 1998 to 2007 were used to train and test the neural network. Prediction accuracy of each network structure was tested using mean square error (MSE), root mean square error (RMSE), and efficiency criterion (R^2). Results showed that the artificial neural network (ANN) predicted groundwater level up to ten weeks ahead with reasonable errors. The accuracy of the network decreases rapidly after that limit. The maximum root mean square error was 0.328 metre

and 0.193 metre for ten-week and one-week lead prediction respectively. As the one-week lead prediction was found almost similar to the actual field value, this could be useful in missing value analysis.

Keywords: Bangladesh; artificial neural network; ANN; groundwater-level prediction; Levenberg-Marquardt algorithm; feedforward backpropagation neural network; FBPNN.

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Biographical notes: Noor-E-Ashmaul Husna is a Research Associate in the Institute of Water and Flood Management at Bangladesh University of Engineering and Technology, Dhaka, Bangladesh. She is currently working on migration as an adaptation option due to climate change for Coastal Belt of Bangladesh. Her general working preferences are GIS and remote sensing application, hydrology, climate change and adaptation, disaster-induced vulnerability analysis, etc.

Sheikh Hefzul Bari is currently working as a Senior Lecturer at Department of Civil Engineering in Leading University, Sylhet. His research interests are statistical hydrology, climate change analysis, GIS and remote sensing application in environmental study, water resources, environmental science etc.

Md. Manjurul Hussain graduated in Civil and Environmental Engineering from Shahjalal University of Science and Technology, Sylhet, Bangladesh. Now, he is pursuing his MSc in Water Resources Development at Institution of Water and Flood Management (IWFM) of Bangladesh University of Engineering and Technology, Dhaka, Bangladesh. He is interested in statistical analysis, GIS, remote sensing and climate change.

Md. Tauhid-Ur-Rahman is a Professor of Civil Engineering faculty at Military Institute of Science and Technology, Dhaka, Bangladesh. His research interests are ground water, climate change etc.

Mashrekur Rahman is a Research Associate in Institute of Water and Flood Management (IWFM) at Bangladesh University of Engineering and Technology (BUET), Dhaka, Bangladesh. He gained his BSc in Civil Engineering in 2014 and currently studying to obtain his MSc in Water Resources Development from IWFM, BUET. His research interests are remote sensing, GIS, salinity modelling, hydrodynamic modelling, climate change studies, constructed treatment wetlands and wastewater treatment methods.

1 Introduction

An optimal groundwater resources management and development strategy often requires continuous forecasting of underground water levels. However, lack of appropriate technology is a constraint for prediction of water level fluctuations in developing countries. Although there is variety of conceptual and process-based simulation techniques available for ground water flow simulation, their application for water level

prediction is limited in the country. Process-based models require an enormous amount of data, which is either expensive or difficult to obtain (Nikolos et al., 2008). On the other hand, Empirical models remain as a suitable alternative method for accurate prediction and can provide reliable results without costly calibration time (Daliakopoulos et al., 2005). ANN is such a model which has an adapting capability to recurrent changes and can detect patterns in a complex natural system (Daliakopoulos et al., 2005).

ANN approach has been exercised globally in the past to forecast the groundwater level (Mao et al., 2002; Nayak et al., 2006; Krishna et al., 2008; Nikolos et al., 2008; Yoon et al., 2011; Rakhshandehroo et al., 2012; Sahoo and Madan, 2013). In Bangladesh, this method is yet to be used widely. High population density and increasing development activity are raising water demand in the country. As a consequence, underground water levels are attenuating, increasing the overall water stress. Therefore, accurate prediction of ground water level is necessary for effective, and sustainable national water resources management policy for the water stressed areas. In addition to management issues, short term prediction would be useful to fill missing data also. Therefore, in this study, a feedforward backpropagation neural network (FBPNN) using Levenberg-Marquardt (LM) algorithm has been employed to predict water level fluctuation at Chandpur District of Bangladesh.

2 Study area

Chandpur District has an area of 1,704.06 sq. km. It is bounded by Comilla District on the east, Meghna River, Shariatpur, Munshiganj districts on the west, Munshiganj and Comilla districts on the north, Noakhali, Lakshmipur and Barisal districts on the south. River erosion is a common feature in this district. Annual average temperature-maximum 34.3°C, minimum 12.7°C; annual average rainfall is 2,551 mm (BBS, 2013). Chandpur city itself stands on the bank of the Meghna River and located at 23°29'–23°40' N 90°6'–90°9' E. Owing to the district's proximity to the coastal belt, salinity is now a prevalent issue. Arsenic and Iron contamination of underground aquifers are also major concerns in that area. The gradual depletion of groundwater is accelerating these problems in this district. The geographical locations of the study area along with the well location are shown in Figure 1.

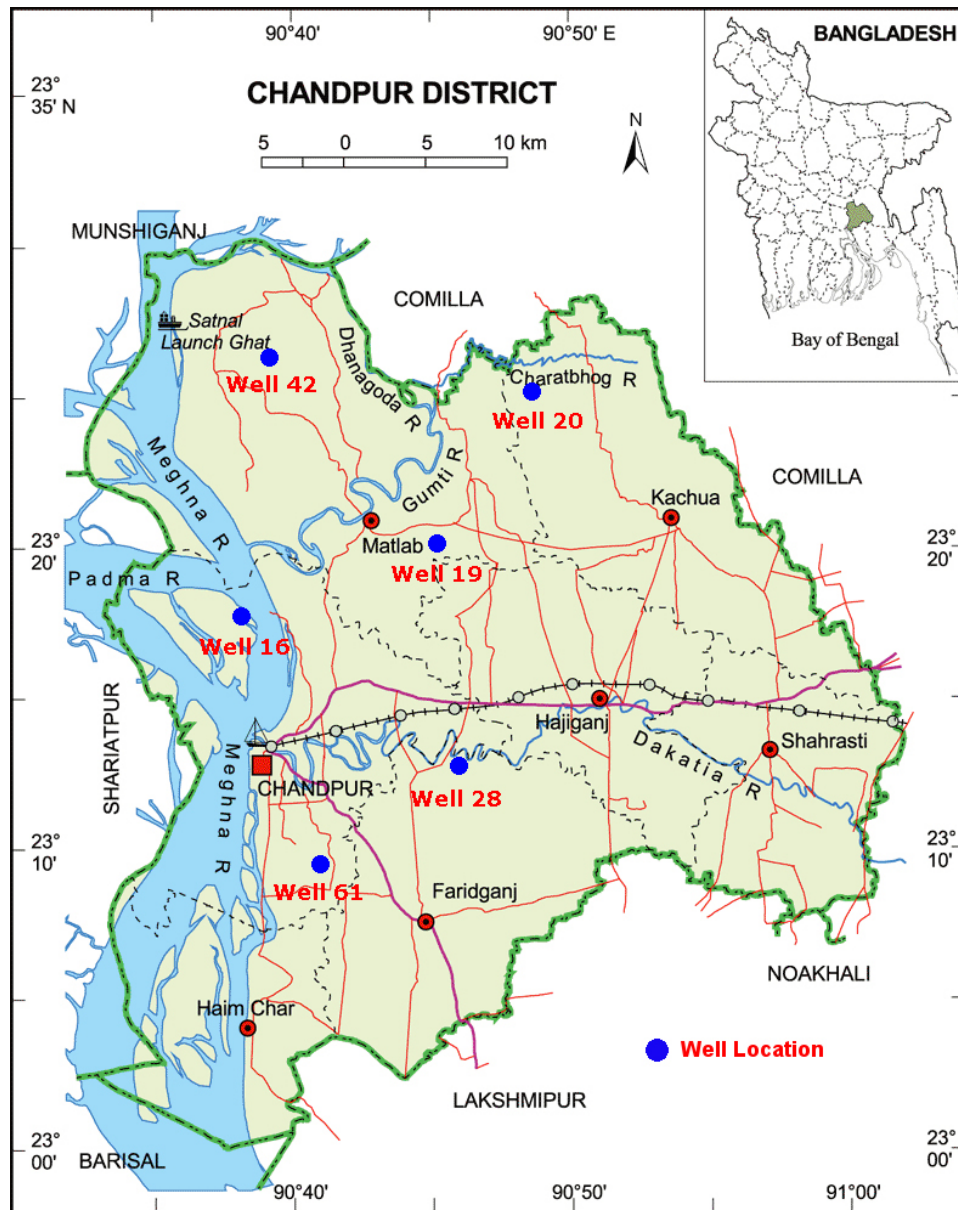
3 Materials and methods

Weekly groundwater level time series data for the years 1998 to 2007 were collected from Bangladesh Water Development Board (BWDB). The data sets were then checked for missing values and probable anomalies. Observation wells having less than 2% missing values were included in the study. Six wells remain for the analysis after checking for missing values. These data were then used for development, training and validation of the artificial neural network (ANN) model.

ANN is a sophisticated soft-computing method comprising of various available procedures. Reviews of past studies revealed that a feedforward back propagation neural network with Levenberg-Marquardt algorithm (LMA) is one of the most plausible options for water level prediction (Daliakopoulos et al., 2005; Rakhshandehroo et al., 2012; Sahoo and Madan, 2013; Yang et al., 2009). Due to the random characteristics of

ANNs, numbers of backpropagation feedforward neural network (BPFNN) models were tested changing number of neurons both in input and hidden layers. A brief description of BPFNN and network terminologies is given in next sections.

Figure 1 Study area with well locations (see online version for colours)



The feedforward network is the simplest ANN network. All layers of this network are connected directly. A feed forward neural network (FFNN) generally uses sigmoid function as a transfer function in the hidden layer and a linear function in the output layer. If the output of FFNN varies from the intended target, the error is minimised by

back propagation algorithm. This error minimising process is called training the network. When the network is fully trained, it may be used to predict future values. To train the FFNN, The LMA (Affandi and Watanabe, 2007; Patel and Jha, 2014) was used, which is faster and produces comparatively minor errors.

The backpropagation algorithm uses the Delta Rule (Widrow and Hoff, 1960), calculating error at output units. After each training case, the effects of error in the output node(s) are propagated backward through the network.

A multi-layer FBPNN is composed of:

- 1 an input layer
- 2 one or more intermediate (hidden) layers
- 3 an output layer.

The output layer may consist of one or more nodes, depending on the problem to be handled. However, it is important to recognize that the term ‘multi-layer’ is often used to refer to multiple layers of weights. This differs with the typical sense of ‘layer’, which refers to a row of nodes (Vemuri, 1994). For clarity, it is often best practice to label an individual network by its number of layers, and the number of nodes in each layer (e.g., a ‘3-4-5’ network has an input layer with three nodes, a hidden layer with four nodes, and an output layer with five nodes) (Leverington, 2009). This structure of network numbering has been used later in this paper.

A smooth, non-linear activation function is vital for a multi-layer network. An activation function usually used in backpropagation networks is the sigma (or sigmoid) function defined as following (Leverington, 2009):

$$a_{jm} = \frac{1}{(1 + e^{-S_{jm}})}$$

where

$$S_{jm} = \sum_{x=0}^n w_{ijx} a_{ix}$$

where a_{jm} is the activation of a particular ‘receiving’ node m in layer j , S_j is the sum of the products of the activations of all relevant ‘emitting’ nodes (i.e., the nodes in the preceding layer (i) by their respective weights and w_{ij} is the set of all weights between layers i and j that are associated with vectors that feed into node m of layer j . This function maps all sums into $[0,1]$ (if the sum of the products is 0, the sigma function returns 0.5, as the sum gets larger, the sigma function returns values closer to 1, and while the function returns values closer to 0 as the sum gets increasingly negative) (Leverington, 2009). The derivative of the sigma function with respect to S_{jm} is conveniently simple, and is given by Gallant (1993) as:

$$\begin{aligned} \frac{d}{d(S_{jm})} (1 + e^{S_{jm}})^{-1} &= -1(1 + e^{-S_{jm}})^{-2} e^{-S_{jm}} (-1) \\ &= \frac{1}{1 + e^{-S_{jm}}} \left(1 - \frac{1}{1 + e^{-S_{jm}}} \right) \\ &= a_{jm} (1 - a_{jm}) \end{aligned}$$

The sigma function applies to all nodes in the network, except the input nodes, whose values are assigned input values. Note that the derivative of the sigma function reaches its maximum at 0.5, and approaches its minimum with values approaching 0 or 1. Thus, the greatest change in weights will occur with values near 0.5 while the least change will occur with values near 0 or 1 (Leverington, 2009).

The accuracy of each ANN models was checked using standard error evaluation procedure. The error evaluation techniques, used in measuring prediction precision are namely, root mean square error (RMSE), mean square error (MSE) and the Efficiency (R^2). These test statistics are widely used in ANN models by various researchers (Affandi and Watanabe, 2007; Yang et al., 2009; Rakhshandehroo et al., 2012). Mathematical expressions for different statistics used in measuring prediction precision are given below:

$$\text{Mean Square Error (MSE)} = \frac{\sum_{i=0}^n (t_i - o_i)^2}{n}$$

$$\text{Root Mean SquareError (RMSE)} = \sqrt{\frac{\sum_{i=0}^n (t_i - o_i)^2}{n}} = \sqrt{\text{MSE}}$$

$$\text{Efficiency Criterion (R}^2\text{)} = 1 - \frac{\sum_{i=0}^n (t_i - o_i)^2}{\sum_{i=0}^n t_i^2 - \frac{\sum_{i=0}^n o_i^2}{n}}$$

where

t_i predicted value

o_i observed value

n number of observed value.

4 Result and discussion

The target of this paper was to predict ground water level for different time intervals. However, the data count in this study was considerably limited and also, there were missing values. The maximum consecutive non-missing value found in the testing period was ten. By running one-week lead, five-week lead, ten-week lead, 15-week lead, etc. models, it was found that one-week lead to ten-week lead models gives reasonable results. After ten-week lead predictions, the error increases rapidly leading towards an unrealistic prediction. Therefore, it could be concluded that up to ten-week lead predictions, FNN is suitable for forecasting groundwater level in Chandpur District. Failure to predict for longer time length may be due to the lack of long span data and a considerable number of missing values within the data set.

Figure 2 Comparison of observed data and predicted data, (a) well 16 (one-week lead) (b) well 16 (ten-week lead) (c) well 19 (one-week lead) (d) well 19 (ten-week lead) (e) well 20 (one-week lead) (f) well 20 (ten-week lead) (g) well 28 (one-week lead) (h) well 28 (ten-week lead) (i) well 42 (one-week lead) (j) well 42 (ten-week lead) (k) well 61 (one-week lead) (l) well 61 (ten-week lead) (see online version for colours)

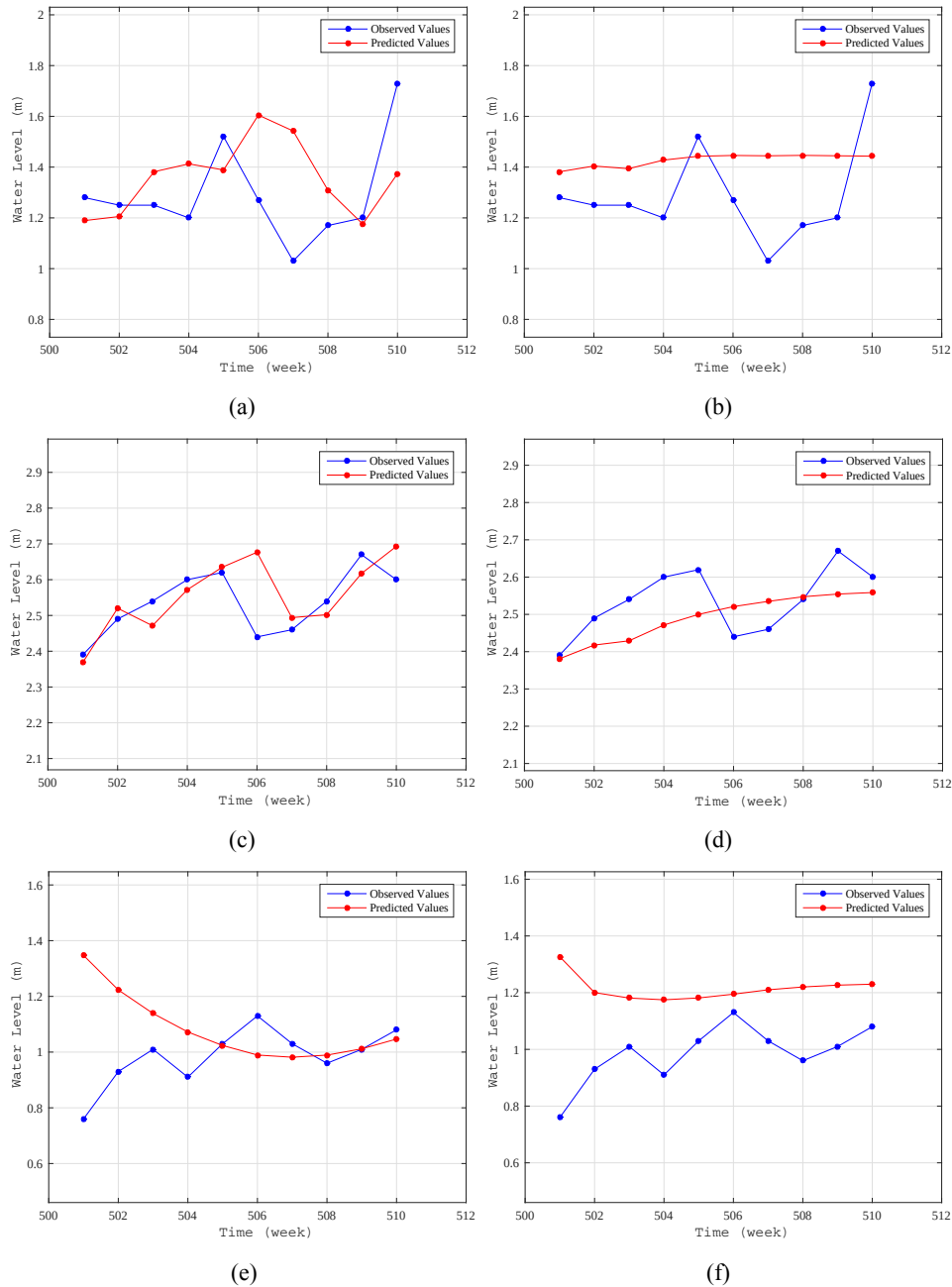
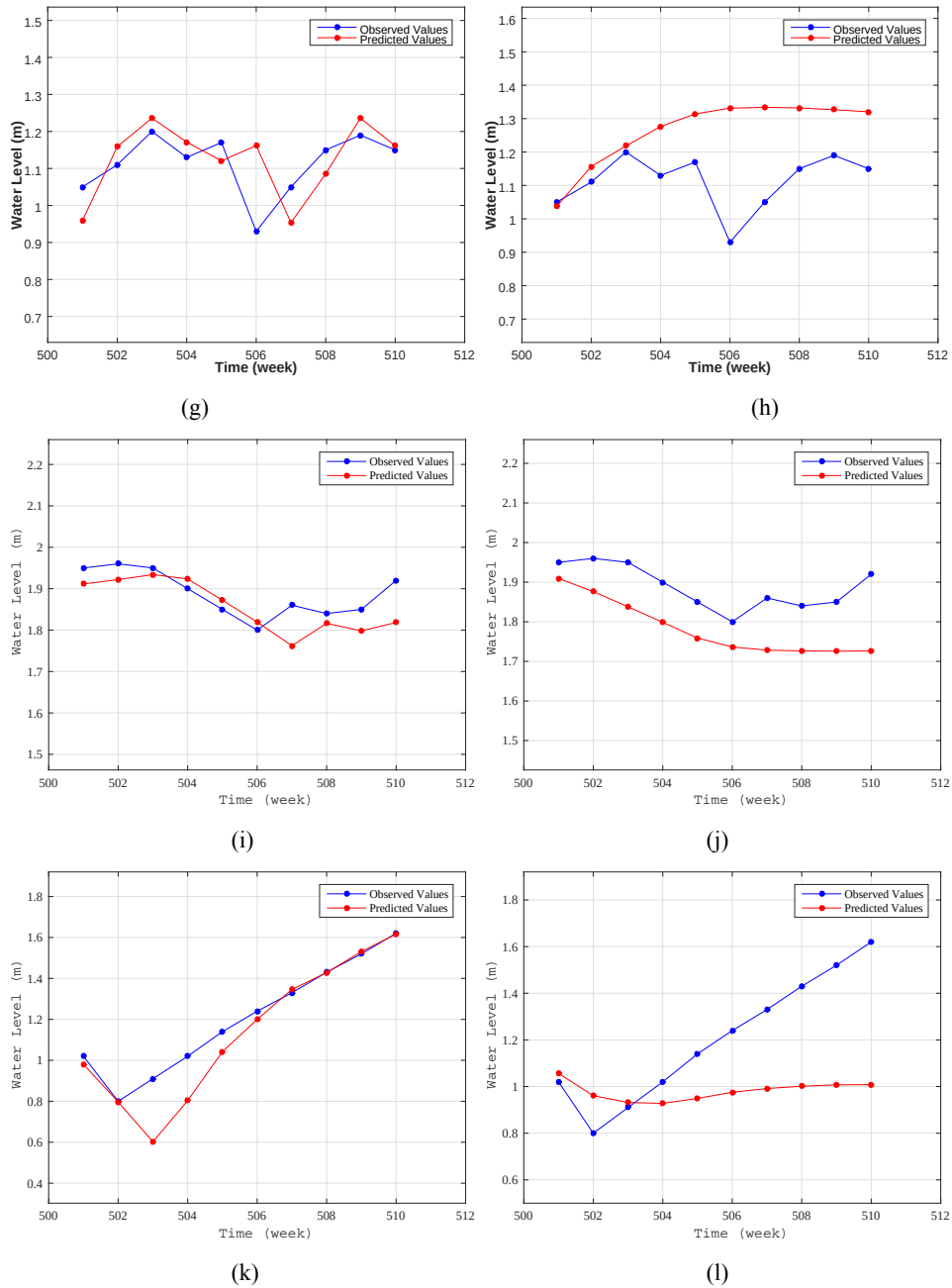


Figure 2 Comparison of observed data and predicted data, (a) well 16 (one-week lead) (b) well 16 (ten-week lead) (c) well 19 (one-week lead) (d) well 19 (ten-week lead) (e) well 20 (one-week lead) (f) well 20 (ten-week lead) (g) well 28 (one-week lead) (h) well 28 (ten-week lead) (i) well 42 (one-week lead) (j) well 42 (ten-week lead) (k) well 61 (one-week lead) (l) well 61 (ten-week lead) (continued) (see online version for colours)



Feedforward back propagation neural network algorithm for a univariate time series was used in this paper. Best fitted FNN models were tested by changing their neurons. Prediction of each point was based on the previous five values. After training and testing for prediction error, it was evident that one-week lead and ten-week lead feedforward neural network (FNN) models give the best results with [5 7 1] and [5 1 5 3 1] network structure respectively. Using these developed networks, prediction of groundwater level was carried out for six geographically distinct wells in Chandpur District. The comparison between actual and predicted values for one-week lead and ten-week lead predictions are shown in Figure 2.

In well 16, the one-week lead prediction is almost the same. Some predictions, like at time 506, 507 have deviated because of abrupt changes in previously observed data. For ten-week lead predictions, the prediction curve moves uniformly for regular values of previous data. Both the graphs for well 20 shows deviation in the first point (time 501). This is due to the high values of previous five data, as each point is predicted-based previous five values. However, the real data (at time 501) suddenly falls down causing higher differences for 501.

RMSE, MSE and the efficiency (R^2) of predicted and actual groundwater depths for all six wells in the one-week lead and ten-week lead models exhibits reasonable results. It is unquestionable that one-week lead prediction gives more accurate result than the ten-week lead prediction. As the one-week lead model predicts the next value only, the error is significantly small there. This type of prediction will be very useful for the calculation of certain epoch's missing values. If it is required to estimate the groundwater level fluctuations for the next few months, then ten-week lead prediction model will smoothly endow the requirement. Although the error in the ten-week lead model is slightly higher, it is not significant and within the acceptable limit as found in the literature (Emamgholizadeh et al., 2014).

Table 1 One-week lead error result

Well ID	RMSE (m)	MSE (m^2)	R^2
16	0.193	0.037	0.960
19	0.024	0.001	0.999
20	0.157	0.025	0.942
28	0.027	0.001	0.992
42	0.009	0	0.999
61	0.049	0.002	0.989

Table 2 Ten-week lead error result

Well ID	RMSE (m)	MSE (m^2)	R^2
16	0.231	0.053	0.964
19	0.087	0.008	0.999
20	0.263	0.069	0.917
28	0.192	0.037	0.966
42	0.113	0.013	0.996
61	0.328	0.108	0.923

Prediction accuracy may depend on the standard deviation of observed data. Data with lower standard deviation can predict more accurately. The prediction for well 19 was the most accurate of all the wells and for well 20 the prediction was least accurate among both models. However, from the results, it could be recommended that ANN may be used as an effective tool for groundwater level prediction, even with limited data up to a certain time step.

5 Conclusions

This study has demonstrated the prediction capability of soft computing techniques for future groundwater level estimation for Chandpur District. The groundwater level prediction for this study was done under two scenarios. The first scenario was a one-week lead prediction, but after appropriate trials, a more satisfactory time limit was chosen. From observation of various models, one-week lead to ten-week lead predictions was found to be acceptable with the used data set.

Best prediction model and appropriate network architecture were selected based on RMSE, R^2 and MSE. The number of hidden layers was optimised by trial and error to one hidden layer. The results of various trials showed that the prediction accuracy declines with increasing prediction time step, with the best accuracy for the one-week lead. Furthermore, it was found that predicted values were close to actual field values and approximately 80% fell within the range of ± 0.2 metres.

As the one-week lead prediction was found to be almost same as the actual field value, this type of prediction may eliminate gaps from missing data. Multiple difficulties in field measurements may result in missing data. One-week lead prediction will be most appropriate to smooth this type of missing values. However, the longer prediction may also fill the large missing gaps in a data series although these will generate some errors. The magnitude of errors would decrease if long-term groundwater data were used. Besides of that long term prediction will help to various decisions concerning groundwater utilisation and management. This long term prediction will also demonstrate seasonal trends in groundwater level, and water extraction could be optimised accordingly. The magnitude of artificial groundwater recharge is a difficult issue due to the uncertainty of water level in various seasons. Long term prediction can minimise such difficulties. Lastly, it could be said that water level prediction can ease a number of decisions regarding groundwater extraction (e.g., irrigation), recharge, salinity intrusion, arsenic contamination, contaminant transport and various other issues.

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